

GENERATIVE PROJECT

Image Captioning with CLIP Encoder + GPT-2 Decoder Milestone #1 Report

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Git Repo Link: https://github.com/lequo-neu/IE7615_Generative_Project

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1. Introduction

This project develops an image captioning system that bridges computer vision and natural language generation. The system combines OpenAI's CLIP (Contrastive Language-Image Pre-training) as a visual encoder with GPT-2 as a language decoder. A learned projection head maps CLIP's visual embeddings into GPT-2's input space using prefix conditioning, enabling the language model to generate descriptive captions conditioned on image content.

The primary goals of this project are:

- Extract meaningful visual representations from images using CLIP ViT-B/32
- Tokenize and prepare natural language captions using GPT-2's tokenizer
- Build a reproducible, modular data pipeline suitable for training in subsequent milestones
- Validate the pipeline end-to-end with sample image-caption test runs
- Establish a scalable project structure for Milestones 2 through 4

This Milestone #1 report covers the dataset selection, preprocessing pipeline, environment setup, CLIP embedding extraction, GPT-2 tokenization, and five sample test runs demonstrating the complete image-to-embedding and caption-to-token pipeline.

2. Dataset Selection and Preprocessing

2.1 Dataset Choice: Flickr30k

We selected the Flickr30k dataset as our primary data source. Flickr30k provides 31,783 images, each annotated with 5 human-written captions. It is a standard benchmark in image captioning research alongside MS-COCO. Key advantages of Flickr30k include: (1) the multi-reference caption structure (5 captions per image) enables robust metric computation (BLEU, METEOR, CIDEr, ROUGE-L) in Milestone #3; (2) straightforward access via HuggingFace without separate API registration; and (3) diverse everyday scenes covering people, animals, objects, and activities.

The dataset was loaded from the HuggingFace Hub using the auto-converted parquet branch (revision refs/convert/parquet) to ensure compatibility with datasets>=4.0, which dropped support for legacy loading scripts.

2.2 Preprocessing Pipeline

The preprocessing pipeline applies the following steps:

- Unicode normalization (NFKD) and lowercasing
- Whitespace collapsing and non-alphanumeric character removal
- Caption length filtering: minimum 8 words, maximum 60 words
- One caption randomly selected per image (all 5 references preserved for Milestone #3 metrics)
- Subsample to 3,000 unique images to keep compute feasible on Apple Silicon (M1)

2.3 Data Split

Split	Images	Ratio
Train	2,400	80.0%

Validation	300	10.0%
Test	300	10.0%
Total	3,000	100%

Table 1: Dataset split distribution.

2.4 Caption Length Distribution

After filtering, the caption length distribution has a mean of 13.0 words (std 4.4), with a range of 8 to 44 words. The distribution is right-skewed, typical of natural language captions where most descriptions are concise.

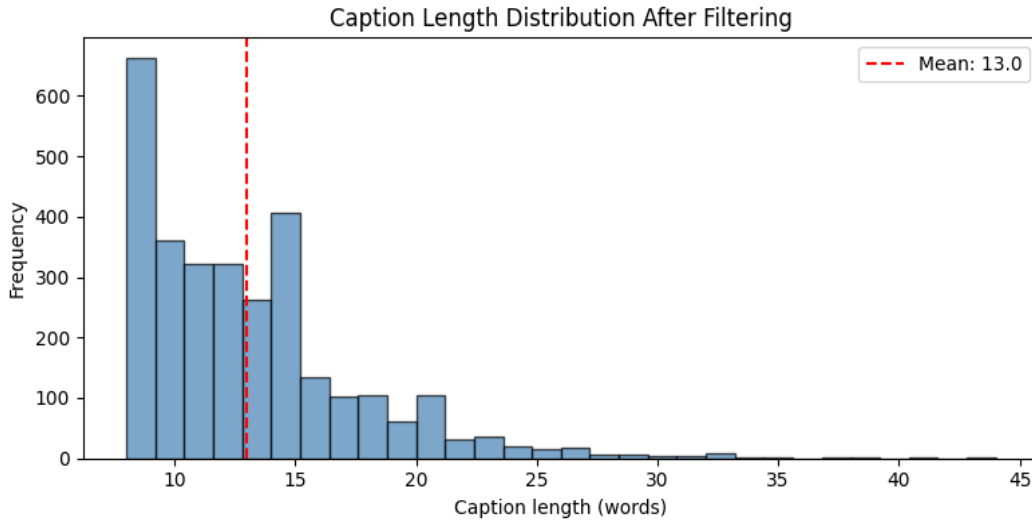


Figure 1: Caption length distribution after filtering.

3. Environment Setup

Package	Version
Python	3.13.7
PyTorch	2.10.0
Transformers	5.2.0
Datasets	4.6.0
NumPy	2.4.2
Matplotlib	3.10.8
Device	MPS (Apple M1)

Table 2: Software environment.

All experiments use a fixed random seed of 42 across Python, NumPy, and PyTorch for reproducibility. Device selection auto-detects MPS (Apple Silicon), CUDA, or CPU. The project uses a centralized configuration dataclass to manage all hyperparameters and paths.

4. Model Architecture Overview

4.1 CLIP Image Encoder

We use `openai/clip-vit-base-patch32` (ViT-B/32) as the image encoder. CLIP is a vision-language model trained on 400M image-text pairs using contrastive learning. The vision transformer processes 224x224 images through 12 transformer layers and produces a 512-dimensional embedding vector. Key properties:

- Total parameters: 151,277,313 (frozen throughout the project)
- Output embedding dimension: 512
- All embeddings are L2-normalized to unit length
- Inference runs on MPS backend with batch size 64

4.2 GPT-2 Tokenizer

Captions are tokenized using the GPT-2 tokenizer (vocabulary size 50,257). Since GPT-2 lacks a native PAD token, we reuse the EOS token (`<|endoftext|>`, id=50256) as the padding token. All captions are padded/truncated to a maximum length of 50 tokens.

4.3 Projection Head (Milestone #2 Preparation)

We define a `ProjectionHead` module that maps CLIP's 512-d embedding space to GPT-2's 768-d input embedding space. The architecture produces a sequence of learned prefix tokens:

Component	Details
Input	(B, 512) CLIP embedding
Layer 1	Linear(512, 7680) + GELU
Layer 2	Linear(7680, 7680) + LayerNorm
Output	(B, 10, 768) prefix tokens
Parameters	62,945,280

Table 3: ProjectionHead architecture.

In Milestone #2, the `ProjectionHead` will be trained to map visual embeddings into prefix tokens that are prepended to GPT-2's input, conditioning the language model on image content.

5. Pipeline Validation Results

5.1 Sample Test Runs

We validated the end-to-end pipeline on 5 test images. For each image, we extracted the CLIP embedding, tokenized the ground-truth caption with GPT-2, and verified the roundtrip decode. All 5 samples passed successfully:

Sample	Caption (truncated)	Tokens	Norm	Roundtrip
0	a shirtless man is climbing up a mountainside.	11 / 50	1.0000	True
75	four friends sit on a grassy hill by a beach...	17 / 50	1.0000	True
150	a girl in red short and a white t-shirt...	18 / 50	1.0000	True
225	a old man in a green shirt sits on the curb...	16 / 50	1.0000	True
299	man standing on rocky ledge overlooking a bay.	10 / 50	1.0000	True

Table 4: Sample test run results.



Figure 2: Five sample images from the test set with ground-truth captions.

5.2 Embedding Space Analysis

We computed pairwise cosine similarity across 50 test images to verify that CLIP embeddings capture meaningful semantic structure. The off-diagonal similarity statistics confirm that embeddings are well-distributed with moderate inter-image similarity, indicating diverse visual representations:

Statistic	Value
Mean	0.5220
Std	0.0877
Min	0.1421
Max	0.8542

Table 5: Off-diagonal cosine similarity statistics (50 test images).

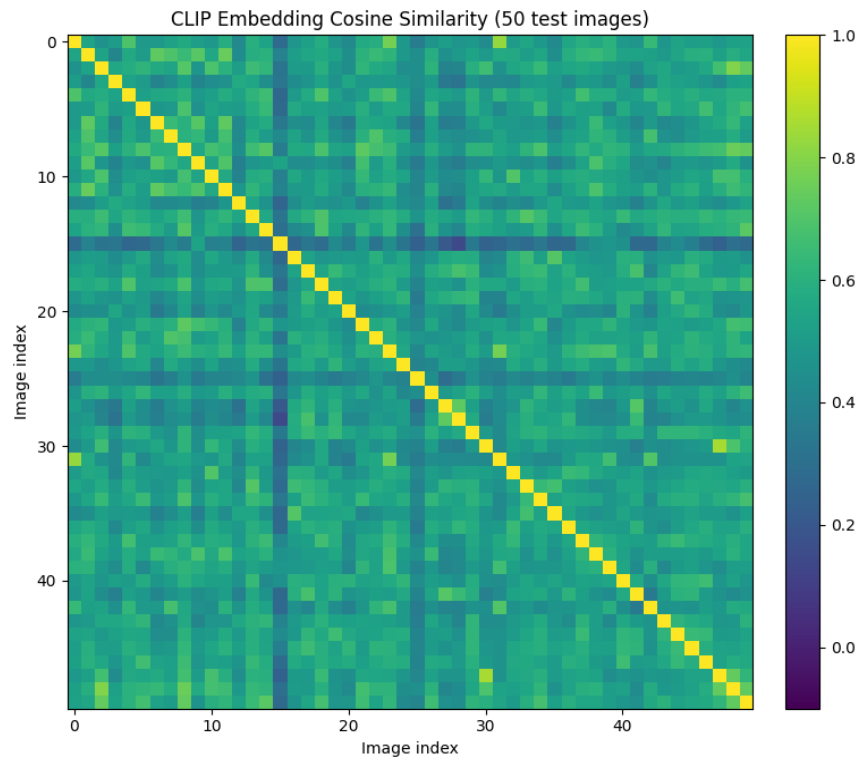


Figure 3: CLIP embedding cosine similarity heatmap (50 test images).

6. Pipeline Component Summary

Component	Status	Output Shape
Dataset loading and cleaning	Complete	3,000 image-caption pairs
Train / Val / Test split	Complete	2400 / 300 / 300
CLIP image encoder (ViT-B/32)	Verified	(B, 512)
GPT-2 tokenizer	Verified	(B, 50)
ProjectionHead (M2 prep)	Defined	(B, 10, 768)
Precomputed embeddings	Saved to disk	.pt files (5.5 + 0.7 + 0.7 MB)
PyTorch DataLoader	Tested	Batch-ready (B=16)

Table 6: Milestone #1 pipeline component status.

7. Team Member Roles

Member	Responsibilities
Quoc Hung Le	Dataset pipeline, CLIP encoder integration, notebook development, report writing
Hassan Alfareed	GPT-2 tokenizer integration, data preprocessing, embedding validation
Khoa Tran	ProjectionHead design, DataLoader testing, repository structure, environment setup

Table 7: Team member role assignments.

8. Milestone #2 Integration Plan

The data pipeline and embedding infrastructure established in Milestone #1 directly support Milestone #2 model training. The planned integration steps are:

1. Load precomputed CLIP embeddings from disk (no recomputation needed)
2. Initialize GPT-2 decoder via `GPT2LMHeadModel.from_pretrained('gpt2')`
3. Train the ProjectionHead to map CLIP embeddings into GPT-2 prefix tokens
4. Implement prefix conditioning: prepend projected embeddings to GPT-2 input
5. Experiment with decoding strategies: greedy, beam search (width 3-5), nucleus sampling (top-p 0.9)
6. Record training logs and generate first caption samples for 5-10 images

The Config dataclass will be extended with training hyperparameters (learning rate, epochs, warmup steps, prefix length) to maintain a single source of truth for all experiment parameters.

9. Repository Structure

- `README.md` — Project overview, setup instructions, milestone tracker
- `environment.yml` — Conda environment for macOS M1
- `requirements.txt` — pip fallback dependencies

- `configs/default.yaml` — Centralized hyperparameters
- `data/raw/` — Original downloads (gitignored)
- `data/processed/` — Cleaned image-caption pairs
- `data/embeddings/` — Precomputed CLIP embeddings (.pt files)
- `notebooks/` — Milestone notebooks (M1 through M4)
- `src/data/` — Dataset and preprocessing modules
- `src/models/` — CLIP encoder, ProjectionHead, GPT-2 decoder
- `src/training/` — Training loop and scheduling
- `src/evaluation/` — Metrics (BLEU, METEOR, CIDEr, ROUGE-L)
- `src/utils/` — Config, seed, logging utilities
- `outputs/` — Logs, checkpoints, figures, samples
- `reports/` — Milestone reports (PDF)